

Fuzzy Linguistic Summaries for Load Forecast Error Diagnostics

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What Problem are we trying to solve

Energy Scheduler

- Energy traders, energy dispatchers
- 20 years of domain experience
- Set reserve margins, hour by hour



Photo: National Laboratory of the Rockies

High responsibility, high risk.

Fast changing environments

- Rapid development of PV production and heatpumps
- Sparse data availability
- Aggregates hide *conditional* structure

The question

Not how large the error is, but *where, when, which direction*.

Why use words

- Operators reason in words: cold, morning, holiday
- Computing with Words makes those labels computable
- LIME and SHAP give numbers

Contributions to adaptive systems

- Adaptive systems in high-risk environments require physical grounding
- Sparse data problems require quick detection of regime changes
- Fuzzy error diagnostics allow for quicker evolution of the system

Method

Data

- CKW smart meters, hourly kWh
- March 2024 – February 2026
- ECMWF temperature, issued **48 h ahead**
- Calendar, solar, lags to 168 h

Model

- Recursive LightGBM, 48 h horizon
- Walk-forward, 363 daily windows

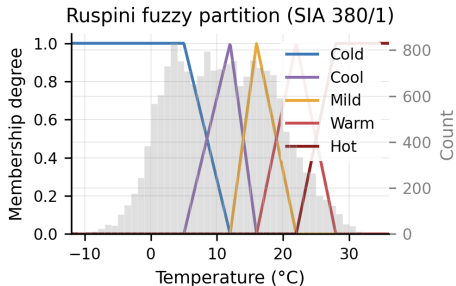
Fuzzy temperature partition

Five terms, breakpoints from the Swiss **SIA 380/1** heating standard.

Ruspini-normalized:

$$\sum_{k=1}^K \mu_k(t) = 1$$

Every hour carries unit weight, so proportions stay undistorted.



Three configurations

- **Baseline:** no fuzzy sets 14 features
- **+Gated Lags:** $\mu_k, y_{t-k} \times \mu$ 35 features
- **+Transitions:** $+\Delta\mu$ 37 features

$\Delta\mu_{\text{cold}} = \mu_{\text{cold}}(t) - \mu_{\text{cold}}(t-1)$, where a positive value means cooling.

"Q of observations where R have S"

$$v = \mu_Q \left(\frac{\sum_i \mu_S(x_i) \wedge \mu_R(x_i)}{\sum_i \mu_R(x_i)} \right)$$

- Quantifiers Q : *Few, About half, Most*
- Each hour contributes in proportion to its membership
- Support reported as $n_{\text{eff}} = \sum_i \mu_R(x_i)$

Results

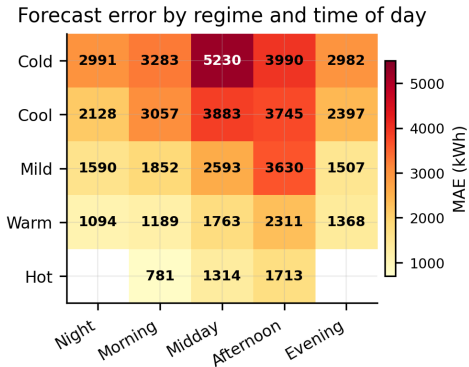
A monotonic error gradient

Regime	n	MAE	% High
Cold	3,608	3,456	34%
Cool	2,021	2,987	28%
Mild	1,793	2,178	15%
Warm	1,011	1,688	10%
Hot	302	1,407	5%

Hot biases +384 kWh, all others -371 to -698. **The sign flips.**

Two restrictions: regime $\times$ time of day

- Cold/midday is worst:
5,230 kWh
- Afternoon peaks in
every regime
- Invisible at regime level
alone



The summaries

Summary	v	n_{eff}
Few Cold hours have Low error	0.58	3,601
Few Warm hours have High error	0.97	1,056
Few Hot hours have High error	1.00	340
Few Cold/Midday hours have Low error	0.95	457
Ab. half Warm/Morning have Low error	0.92	187
Ab. half Cold weekend over-predict	0.96	1,026
Ab. half Cold weekday under-predict	0.63	2,575

7 of 18 summaries; full table in appendix.

Where transitions help, and hurt

Subdomain	Impr.
Mild / Morning	+19.0%
Cool / Morning	+15.3%
Cool / Midday	+13.2%
Cold / Midday	+5.3%
Cold / Evening	-3.9%
Hot / Morning	-17.6%

- Gains in **cool and mild**, the heating crossover
- $\Delta\mu$ informs where the regime actively shifts
- Cold/evening: stable regime, stale signal
- Hot/morning: only $n = 32$

Adaptation should be **modulated by regime stability**.

Errors cluster

- Runs of **3.9 h** in cold
- vs. 2.8 h in mild and warm
- High-error mornings predict high-error middays ($v = 0.98$)

Cold-regime bias (kWh)

	Bias	<i>n</i>
Holiday	−1,073	283
Normal	−322	3,325
Weekend	+250	1,034
Weekday	−635	2,574

What a dispatcher does with it

- Every restriction is known **at prediction time**
- Compute $\mu_k(t)$, set **hour-specific reserve margins**
- Cold/midday widest; cold weekends need the *opposite* sign

2,810 kWh is a number. This is a diagnosis.

Limitations

- Descriptive, not causal
- Hot regime $n = 302$; 16 cold holiday days only
- Wu & Mendel coverage ignores the *number* of restrictions

Next

- Restriction-aware coverage measures
- Fuzzy functional dependencies for coupled time series
- Regime-conditional ensemble weighting from $\mu_k(t)$

github.com/j4yk4y/fuzzy-ls-forecast-diagnostics

Thank you!

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Backup: trapezoidal memberships

$$\mu_k(t) = \max\left(0, \min\left(\frac{t-a}{b-a}, 1, \frac{d-t}{d-c}\right)\right)$$

- *Cold* and *hot* are shoulder functions
- *Cool*, *mild*, *warm* have $b = c$, so they are triangular
- Interactions $t \times \mu_{\text{cold}}$, $t \times \mu_{\text{cool}}$ give regime-specific temperature sensitivity

Backup: quantifier membership functions

Quantifier	$\mu_Q(p)$	$v = 1$
Few	$\max(0, (0.3 - p)/0.2)$	$p \leq 0.1$
About half	$\max(0, 1 - p - 0.5 /0.2)$	$0.3 \leq p \leq 0.7$
Most	$\max(0, \min(1, (p - 0.5)/0.3))$	$p \geq 0.8$

Minimum t-norm for $\wedge$. A proportion can belong to two quantifiers at once; all with $v > 0$ are reported.

Backup: LightGBM hyperparameters

Parameter	Value
Number of leaves	63
Learning rate	0.05
Boosting rounds	500
Min. child samples	20
Row subsample	0.9
Column subsample	0.8
L1 / L2 (α , λ)	0.0
Random seed	271
Device	GPU

Not tuned per configuration; the goal is comparable residuals, not maximum accuracy.

Backup: experimental design

- Restrictions: regime, time of day, calendar, model configuration
- Time blocks: night 0–5, morning 6–9, midday 10–14, afternoon 15–18, evening 19–23
- Errors binned Low/Med/High on **global** quartiles, so “high” means the same in every regime
- Tables defuzzify to the dominant regime; summaries use full memberships
- Diagnostics run on **Baseline** residuals: no fuzzy features, hence no circularity

Backup: all 18 summaries

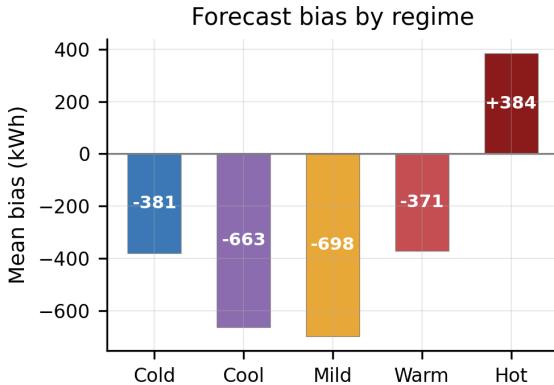
Summary	p	v	n_{eff}
Few Cold hours have Low error	18%	0.58	3,601
Ab. half Cold hours have Med. error	47%	0.87	3,601
Ab. half Cool hours have Med. error	51%	0.95	2,028
Ab. half Mild hours have Med. error	55%	0.75	1,709
Few Mild hours have High error	15%	0.74	1,709
Few Warm hours have High error	11%	0.97	1,056
Ab. half Warm hours have Low error	40%	0.51	1,056
Few Hot hours have High error	5%	1.00	340
Ab. half Hot hours have Med. error	49%	0.93	340
Few Cold/Midday hours have Low error	11%	0.95	457
Ab. half Cold/Night hrs have Med. err	50%	0.99	1,181
Few Mild/Night hours have High error	5%	1.00	397
Few Warm/Midday hours have High error	11%	0.94	352
Ab. half Warm/Morning hrs have Low err	52%	0.92	187
Ab. half Cold weekend hrs over-predict	49%	0.96	1,026
Ab. half Cold weekday hrs under-predict	57%	0.63	2,575
Most Hot weekday hours over-predict	68%	0.60	225
Most Cold days, high error mornings also high error middays	79%	0.98	39 days

Backup: full error table

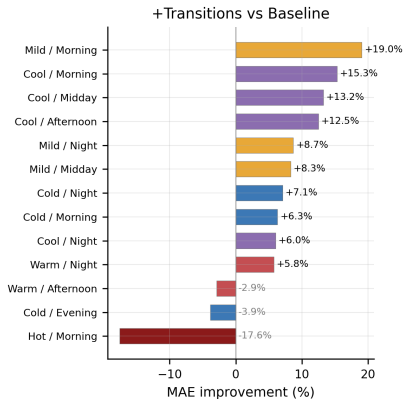
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Mild	1,793	2,178	30%	56%	15%
Warm	1,011	1,688	42%	48%	10%
Hot	302	1,407	46%	49%	5%

Global MAE: Baseline 2,810 kWh, +Gated Lags 2,712, +Transitions 2,657 kWh.

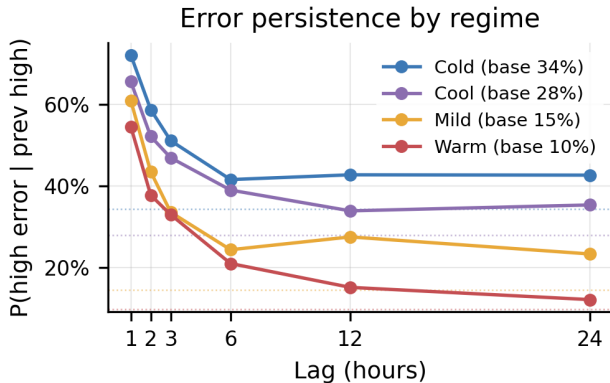
Backup: bias by regime



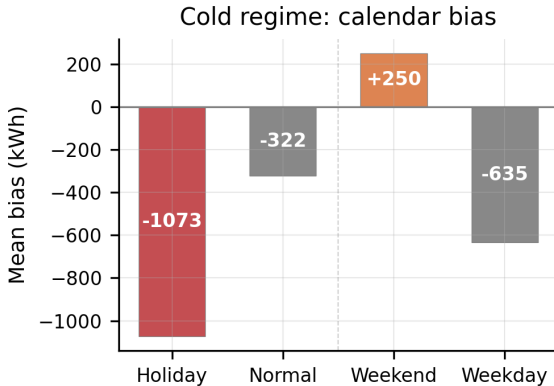
Backup: cross-model improvement



Backup: error persistence



Backup: calendar bias



Backup: monthly MAE

